

# CARLA C. TY

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## SUMMARY

Innovative and detail-oriented Computer Engineering undergraduate with hands-on experience in web development, embedded systems, Image Processing, IoT development, and circuit design. Experience in integrating hardware and software to build efficient, real-world solutions. Gained strong communication, problem-solving, adaptability, and conflict resolution skills through customer service roles, enhancing the ability to work effectively in fast-paced, collaborative environments. Passionate about designing scalable systems and eager to contribute to impactful technology projects while continuing to grow in the tech industry.

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## EXPERIENCE

### CUSTOMER SERVICE REPRESENTATIVE EXPERIENCE (TELECOMMUNICATION ACCOUNT)

#### Alorica

May 2023 - SEPTEMBER 2023, Cubao, Quezon City

- Resolved 30+ customer inquiries daily with professionalism and speed
- Communicated technical details clearly to non-technical users
- Demonstrated reliability by maintaining a 95% schedule adherence rate
- Collaborated with other departments to improve customer support workflows

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## PROJECT

### IMAGE PROCESSING BAMBOO USING YOLO ALGORITHM

IEEEExplore • May 2024 - May 2025

- Develop a device to distinguish between running and clumping bamboo growth habits using image data
- Utilize the YOLOv5 algorithm to automatically detect and classify bamboo growth habits patterns, allowing for efficient processing of image-based datasets
- Analyzed the performance of the created model using a confusion matrix
- Evaluate the practical applicability of the device and algorithm in real-world bamboo habitats.
- Participate and thesis presentation during 2025<sup>th</sup> 7<sup>th</sup> Asia Symposium on Image Processing held in Tsukuba, Japan (June 13 to 15, 2025).

### PAPER PRINTING VENDING MACHINE USING ATMEGA328P

ICpEP - MAPUA • May 2025 - May 2025

- Developed a coin-operated printing vending machine using an ATmega328P microcontroller and Raspberry Pi 4B.
- Implement a combinational circuit (PCB) for using a 7-segment display for a synchronous countdown of pages printed, controlled via a JK flip-flop-based 4-bit down counter.
- Accepts I-peso and 5-peso coins with a coin slot module, dispensing forms priced at 3 pesos each.
- Features a user interface with a keyboard for entering required information on the form before printing.
- Equipped with a sensor and buzzer to alert when paper supply is empty.

### AWSCC-MAPUA OFFICAL WEBSITE (BACKEND DEVELOPER)

[mapua.awscceph.org](http://mapua.awscceph.org) • November 2024 - May 2025

- A collaboration work with other members of AWSCC- MAPUA to make a website for AWS Club Mapua.
- Designed and implemented a serverless web feature using **AWS DynamoDB, S3, and Lambda** to manage officer profiles.
- Developed an **HTML-based Lambda form** for uploading officer photos to **S3** and submitting profile data (full name, position, description, LinkedIn link) to **DynamoDB**.
- Integrated **CRUD operations** via Lambda functions to manage profile data.
- Built and tested the entire system **locally using AWS SAM**, with code organized in a self-contained repository prepared for forking and deployment.
- AWS Service: API Gateway, Lambda, S3, DynamoDB, IAM

### St. Bede Medical Center Website

<https://tycarla.github.io/st.bede-medical-center/> • July 2025

- Designed a professional medical website for a client with a clean, user-friendly interface.
- Built a homepage ( `index.html` ) highlighting services, core values, news, and contact details.
- Created an Appointment Page ( `appointment.html` ) for patients to set up consultations.
- Designed an Electronic Records Management (ERM) interface ( `ERM.html` ) to display patient records.
- Optimized the site for deployment via GitHub Pages.
- Maintained clean and modular HTML structure for each major page section.
- Organized project assets including icons, images, and layout backgrounds.

## **IOT WITH MINI SMART MEDICINE CABINET**

Mapua University

- Integrates sensor technology for effective medication management and storage.
- Temperature sensor ensures optimal storage conditions, maintaining medication potency.
- Ultrasonic sensor tracks cabinet access by logging door openings and closings for usage monitoring and security.
- IR sensor monitors the quantity of medicines, helping users keep track of stock levels.
- Automated reminders through a buzzer and LED indicators alert users to low stock or if the cabinet is left open.
- Prioritizes medication adherence and safety, preventing errors and maintaining proper storage conditions.

## **MINI SMART BICYCLE FOR CYCLIST SAFETY**

Mapua University

- Addresses the need for cyclist protection using ultrasonic sensors, LED indicators, a microcontroller Arduino Uno, LCD display, and a buzzer.
- Alerts cyclists about approaching vehicles from behind with a multi-faceted warning system.
- Incorporates a manual turning signal system for enhanced situational awareness.
- Features three buttons for signaling: left, right, and off.
  - Pressing left or right triggers corresponding blinking yellow LEDs to indicate direction.
  - The middle button turns off the signal.
- LCD display shows safety recommendations and messages to the cyclist.

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## **EDUCATION**

### **BACHELOR OF SCIENCE IN COMPUTER ENGINEERING**

Specialization in EMBEDDED SYSTEM TECHNOLOGY (ONGOING) • Mapua University • Intramuros Manila • AUGUST 2020 - DECEMBER 2025

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## **CERTIFICATIONS**

### **ARTIFICIAL INTELLIGENCE TECHNOLOGY AND APPLICATIONS HUAWEI ISSUED**

HUAWEI • JUNE 2025

### **CCNA: ENTERPRISE NETWORKING, SECURITY, AND AUTOMATION CISCO ISSUED**

CISCO • DECEMBER 2024

### **NETWORK SECURITY CISCO ISSUED**

CISCO • NOVEMBER 2024

### **CCNA: INTRODUCTION TO NETWORKS CISCO ISSUED**

CISCO • FEBRUARY 2024

### **INTRODUCTION TO EMBEDDED SYSTEMS SOFTWARE AND DEVELOPMENT ENVIRONMENTS**

COURSERA • 2025

### **OPERATING SYSTEMS AND YOU: BECOMING A POWER USER COURSERA ISSUED**

COURSERA • APRIL 2022

### **SYSTEM ADMINISTRATION AND IT INFRASTRUCTURE SERVICES COURSERA ISSUED**

COURSERA • OCTOBER 2022

### **THE BITS AND BYTES OF COMPUTER NETWORKING COURSERA ISSUED**

COURSERA • MAY 2021

### **TECHNICAL SUPPORT FUNDAMENTALS GOOGLE ISSUED**

GOOGLE • MAR 2021

### **INTRODUCTION TO GIT AND GITHUB COURSERA ISSUED**

COURSERA • MAY 2021

### **APPLIED PLOTTING, CHARTING & DATA REPRESENTATION IN PYTHON COURSERA ISSUED**

COURSERA • OCTOBER 2022

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## **SKILLS**

**Microcontrollers & Development Boards:** Arduino UNO, ESP32, ESP8266, Raspberry Pi 4, ATMEGA328P

**Programming Language:** C/C++, Embedded C, Python, JavaScript

**ECAD:** Proteus8, KiCad, Tinkercad

**PCB Assembly:** PCB Design, Circuit Schematic Design, Sensor Integration

**Communication Protocols:** USB-UART, Bluetooth Classic (BR/EDR) & Bluetooth Low Energy (BLE), Wi-Fi 802.11 b/g/n

**Version Control & Development Tools:** Git/GitHub, VS Code, Anaconda, Jupyter Notebook, Docker

**Networking & Cloud Integration:** AWS IoT, HTTP/HTTPS APIs, RESTful Services, DynamoDB, AWS Lambda, AWS IAM, AWS S3, RTOS Basics